ATTENDIFY PRO – STUDENT ATTENDANCE TRACKER

1. Introduction

Attendify Pro is a web-based application used to manage student attendance. It helps teachers add students, mark attendance, view attendance statistics, and download attendance reports.

The main aim of this project is to make attendance management simple, quick, and easy.

2. Problem Statement

Maintaining attendance manually takes time and effort. Calculating attendance percentages and identifying students with low attendance can also be difficult.

Our project provides a digital solution to make these tasks easier.

3. Objectives

* To simplify student attendance management.
* To add and manage student details.
* To record daily attendance.
* To calculate attendance statistics.
* To identify students with attendance below 75%.
* To download attendance reports.

4. Technologies Used

* HTML: Creates the structure of the website.

* CSS: Makes the website attractive and user-friendly.

* JavaScript: Handles attendance marking, calculations, and interactive features.

* Browser Storage: Saves student and attendance information in the browser.

* GitHub Pages: Hosts the website online.

Note: The website uses browser-based storage, but the exact storage technology and source-code implementation should be verified before submission.

5. Main Features

1. Dashboard: Displays attendance statistics.

2. Student Management: Allows users to add student names and roll numbers.

3. Attendance Tracking: Allows users to mark students present or absent.

4. Analytics: Displays attendance information and identifies students with low attendance.

5. CSV Export: Allows users to download attendance records.

6. Settings: Provides options to reset attendance or clear stored data.

6. Working of the Project

The application works through the following steps:

1. Open the Attendify Pro website.

2. Add students using their names and roll numbers.

3. Select the attendance date.

4. Mark students as present or absent.

5. View the attendance statistics on the dashboard.

6. Check the analytics section.

7. Export the attendance records if required.

Working Flow

Open Website → Add Students → Mark Attendance → View Statistics → Export Report

7. Attendance Calculation

The attendance percentage can be calculated using the following formula:

Attendance Percentage = (Number of Present Sessions / Total Marked Sessions) × 100

Example:

If a student attends 18 out of 20 sessions:

Attendance Percentage = (18 / 20) × 100 = 90%

The application helps users understand attendance levels and identify students whose attendance is below 75%.

8. Advantages

* Easy to use.
* Reduces manual work.
* Simplifies attendance calculations.
* Provides a clear attendance summary.
* Allows attendance records to be downloaded.
* Helps identify students with low attendance.

9. Limitations

* Attendance data is stored in the browser.
* Data may be lost if the stored browser data is cleared.
* Records may not automatically synchronize between devices.
* Multiple teachers cannot be assumed to share the same records.

10. Future Enhancements

* Add a secure login system.
* Integrate a centralized database.
* Provide cloud backup.
* Allow multiple teachers to access shared records.
* Generate PDF attendance reports.
* Send notifications for students with low attendance.

11. Conclusion

Attendify Pro is a simple attendance management application that makes it easier to manage student records and track daily attendance.

It provides features such as student management, attendance tracking, analytics, and report export.

The project demonstrates how web technologies can be used to solve a common problem in educational institutions.

12. Reference

Project Website: https://diviyatech.github.io/attendance-tracker/